

Yi Gao

Robot Learning · Contact-Rich Manipulation · Humanoid Compliance

gaoe05@qq.com | GaoE05.github.io | github.com/GaoE05 | Reading notes (Xiaohongshu)

Available immediately for remote research collaboration: 30 hours/week

Education

SWUFE-UD Institute of Data Science

Sep 2024 - Jun 2028 (expected)

University of Delaware · B.S. in Information Systems

Southwestern University of Finance and Economics · Information Management and Information Systems

China-based dual-degree program · GPA: **87.7/100**

Research Experience

SymBiosis Robotics

May 2026 - Present

Research Intern; ongoing remote collaboration · Advisor: Pengxiang Ding

SoftSONIC: Residual Learning for Humanoid Compliance

- **Independently led SoftSONIC** from method design and data generation to RL experiments and deployment validation, with research guidance from senior collaborators. Studied lightweight compliance adaptation of an existing humanoid motion prior.
- Developed a bounded **64-D latent residual** over the frozen SONIC encoder and decoder; trained it with PPO and SoftMimic-inspired compliant references. The deployed actor uses proprioceptive/action history without explicit external-force measurements.
- Trained and evaluated a **shared residual policy on 32 training motions in simulation**, using paired perturbations and no-force tests; compared joint-space residuals, nominal-token conditioning, and reward variants to study compliance, motion preservation, and transfer.
- Implemented and validated a **50 Hz C++/TensorRT deployment** based on official SONIC. Observed qualitative yielding under manual pulls on a **Unitree G1** during standing, walking, and selected motion replays, including a backward-walking reference excluded from residual-policy training. Observations remain preliminary; response strength varies across motions.

Lenovo Robotics Research Institute (Shanghai)

Jul 2025 - Dec 2025

Research Intern · Solution & Development Center

- Developed a context-grounded robot agent prototype for industrial patrol, integrating visual perception, location/time-specific safety rules, and action protocols into a traceable decision pipeline.
- Collected and organized **200+ real-world video clips** using a DEEPRobotics X30 quadruped; built a structured annotation and VLM evaluation workflow comparing QwenVL and GLM-family models.
- Drafted *CE4Patrol: Multi-Layer Context Reasoning for Industrial Anomaly Inspection* (manuscript; Yi Gao, et al.).

Selected Technical Project

VLA Inference for Retail Mobile Manipulation

Mar 2026 - Apr 2026

Reproduced a **RoboBenchMart** VLA inference pipeline in ManiSkill for retail mobile manipulation. Set up assets, scene generation, model serving, and evaluation-client integration for pick-and-place and open/close tasks.

Technical Skills

Programming & tools: Python, C++, PyTorch, Git, Linux, Docker, TensorRT

Simulation: Isaac Lab / Isaac Sim, MuJoCo, ManiSkill, PyBullet

Methods: PPO, residual policy learning, motion tracking, imitation learning, experimental diagnostics

Honors & Languages

Meritorious Winner, Mathematical Contest in Modeling (COMAP), 2025

Second Prize, National English Competition for College Students, 2025 · CET-4: 632; CET-6: 599